BORKO FURHT
Editor

HANDBOOK OF SOCIAL NETWORK TECHNOLOGIES AND APPLICATIONS

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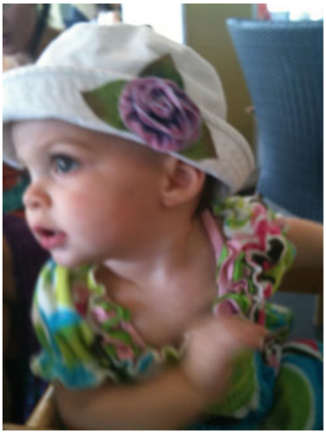

To my first granddaughter Sophia Rolleri, who is the cutest in the world.

Preface

Social networking is a concept that has been around for a long time; however, with the explosion of the Internet, social networking has become a tool for connecting people and allowing their communications in the ways that was previously impossible. Furthermore, the recent development of Web 2.0 has provided for many new applications such as Myspace, Facebook, Linkedin, and many others.

The objective of this Handbook is to provide comprehensive guidelines on the current and future trends in social network technologies and applications. This Handbook is a carefully edited book – contributors are 82 worldwide experts in the field of social networks and their applications. The Handbook Advisory Board, comprised of 11 researchers and practitioners from academia and industry, helped in reshaping the Handbook and selecting the right topics and creative and knowledgeable contributors.

The scope of the book includes leading edge social network technologies, infrastructures, and communities; social media analysis, organizations, mining, and search; privacy and security issues in social networks; and visualization and applications of social networks.

The Handbook comprises of five parts, which consist of 31 chapters. The first part on *Social Media Analysis and Organization* includes chapters dealing with structure and dynamics of social networks, qualitative analysis of commercial social networks, and various topics relating to analysis and organization of social media.

The second part on *Social Media Mining and Search* focuses on chapters on detecting and discovering communities in social networks, and mining information from social networks and related topics in social media mining and search.

The third part on *Social Network Infrastructures and Communities* consists of chapters on various issues relating to distributed and decentralized online social networks, accessibility testing of social Websites, and understanding human behavior in social networks and related topics.

The fourth part on *Privacy in Online Social Networks* describes various issues related to security, privacy threats, and intrusion detection in social networks.

The fifth part on *Visualization and Applications of Social Networks* includes chapters on visualization techniques for social networks as well as chapters on several applications.

With the dramatic growth of social networks and their applications, this Handbook can be the definitive resource for persons working in this field as researchers, scientists, programmers, engineers, and users. The book is intended for a wide variety of people including academicians, designers, developers, educators, engineers, practitioners, researchers, and graduate students. This book can also be beneficial for business managers, entrepreneurs, and investors. The book can have a great potential to be adopted as a textbook in current and new courses on Social Networks.

The main features of this Handbook can be summarized as follows:

1. The Handbook describes and evaluates the current state-of-the-art in the field of social networks.
2. It also presents current trends in social media analysis, mining, and search as well social network infrastructures and communities.
3. Contributors to the Handbook are the leading researchers from academia and practitioners from industry.

We would like to thank the authors for their contributions. Without their expertise and effort, this Handbook would never come to fruition. Springer editors and staff also deserve our sincere recognition for their support throughout the project.

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Editor-in-Chief

Editor-in-Chief

Borko Furht is a professor and chairman of the Department of Electrical & Computer Engineering and Computer Science at Florida Atlantic University (FAU) in Boca Raton, Florida. He is also director of recently formed NSF-sponsored Industry/University Cooperative Research Center on Advanced Knowledge Enablement. Before joining FAU, he was a vice president of research and a senior director of development at Modcomp (Ft. Lauderdale), a computer company of Daimler Benz, Germany, a professor at University of Miami in Coral Gables, Florida, and a senior researcher in the Institute Boris Kidric-Vinca, Yugoslavia. Professor Furht received Ph.D. degree in electrical and computer engineering from the University of Belgrade. His current research is in multimedia systems, video coding and compression, 3D video and image systems, wireless multimedia, and Internet, cloud computing and social networks. He is presently Principal Investigator and Co-PI of several multiyear, multimillion dollar projects including NSF PIRE project and NSF High-Performance Computing Center. He is the author of numerous books and articles in the areas of multimedia, computer architecture, real-time computing, and operating systems. He is a founder and editor-in-chief of *The Journal of Multimedia Tools and Applications* (Springer). He has received several technical and publishing awards and has consulted for many high-tech companies including IBM, Hewlett-Packard, Xerox, General Electric, JPL, NASA, Honeywell, and RCA. He has also served as a consultant to various colleges and universities. He has given many invited talks, keynote lectures, seminars, and tutorials. He served on the Board of Directors of several high-tech companies.

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Part I
Social Media Analysis and Organization

Chapter 1
Social Network Analysis: History, Concepts, and Research

Mingxin Zhang

1.1 Introduction

Social network analysis (SNA), in essence, is not a formal theory in social science, but rather an approach for investigating social structures, which is why SNA is often referred to as structural analysis [1]. The most important difference between social network analysis and the traditional or classic social research approach is that the contexts of the social actor, or the relationships between actors are the first considerations of the former, while the latter focuses on individual properties. A social network is a group of collaborating, and/or competing individuals or entities that are related to each other. It may be presented as a graph, or a multi-graph; each participant in the collaboration or competition is called an actor and depicted as a node in the graph theory. Valued relations between actors are depicted as links, or ties, either directed or undirected, between the corresponding nodes. Actors can be persons, organizations, or groups – any set of related entities. As such, SNA may be used on different levels, ranging from individuals, web pages, families, small groups, to large organizations, parties, and even to nations.

According to the well known SNA researcher Lin Freeman [2], network analysis is based on the intuitive notion that these patterns are important features of the lives of the individuals or social entities who display them; Network analysts believe that how an individual lives, or social entity depends in large part on how that they are tied into the larger web of social connections/structures. Many believe, moreover, that the success or failure of societies and organizations often depends on the patterning of their internal structure.

With a history of more than 70 years, SNA as an interdisciplinary technique developed under many influences, which come from different fields such as sociology, mathematics and computer science, are becoming increasingly important across many disciplines, including sociology, economics, communication science, and psychology around the world. In the current chapter of this book, the author discusses

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the history of social network analysis, including its origin and development, in a brief manner, and discusses definition, features, fundamental concepts and research of social network analysis. In the end of this chapter, the author points out that developing and looking for new techniques and tools that resolve corresponding problems challenging SNS research is urgent.

1.2 Social Network Analysis: Definition and Features

Social network is formally defined as a set of social actors, or nodes, members that are connected by one or more types of relations [3]. Nodes, or network members, are the units that are connected by the relations whose patterns researchers study. The units are most commonly individuals, groups or organizations, but in principle any units that can be connected to other units can be studied as nodes, such as web pages, blogs, emails, instant messages, families, journal articles, neighborhoods, classes, sectors within organizations, positions, or nations [4–6]. Research in a number of academic fields has shown that social networks operate on many levels, from families up to the level of nations, and play a critical role in determining the way problems are solved, organizations are run, and the degree to which individuals succeed in achieving their goals [7].

Traditionally, mainstream social research focus exclusively on the behavior of individuals. This approach neglects the social part or structure of human behavior; the part that is concerned with the ways individuals interact and the influence they have on one another [8]. However, social network analysts, take these parts as the primary building blocks of the social world, they not only collect unique types of data, they begin their analyses from a fundamentally different perspective than that had been not adopted by individualist or attribute-based social science.

In social science, the structural approach, that is based on the study of interaction among social actors is called social network analysis [8]. The relationships that social network analysts study are usually those that link individual human beings, since these social scientists believe that besides individual characteristics, relational links or social structure, are necessary and indispensable to fully understand social phenomena. Specifically, Wetherell et al. describe social network analysis as follows [9, p. 645]:

> *Most broadly*, social network analysis *(1) conceptualizes social structure as a network with ties connecting members and channelling resources, (2) focuses on the characteristics of ties rather than on the characteristics of the individual members, and (3) views communities as 'personal communities', that is, as networks of individual relations that people foster, maintain, and use in the course of their daily lives.*

Structural approach is not confined to the study of human social relationships. As Freeman pointed out [8], it is present in almost every field of science. For example, Freeman wrote that, molecular chemists examine how various kinds of atoms interact together to form different kinds of molecules, while electrical engineers observe how the interactions of various electronic components – like capacitors and resistors – influence the flow of current through a circuit. And biologists study the ways in which each of the species in an ecosystem interacts with and impinges on each of the others.

There are different types of networks. Generally, network analysts differentiate the following networks:

One Mode Versus Two Mode Networks. The former involve relations among a single set of similar actors, while the latter involve relations among two different sets of actors. An example of two mode network would be the analysis of a network consisting of private, for profit organizations and their links to non-profit agencies in a community [10]. Two mode networks are also used to investigate the relationship between a set of actors and a series of events. For example, although people may not have direct ties to each other, they may attend similar events or activities in a community and in doing so, this sets up opportunities for the formation of "weak ties" [11].

Complete/Whole Versus Ego Networks. Complete/whole or Socio-centric networks consist of the connections among members of a single, bounded community. Relational ties among all of the teachers in a high school is an example of whole network. Ego/Ego-centric or personal networks are referred to as the ties directly connecting the focal actor, or ego to others, or ego's alters in the network, plus ego's views on the ties among his or her alters. If we asked a teacher to nominate the people he/she socializes with outside of school, and then asked that teacher to indicate who in that network socializes with the others nominated, it is a typical ego network.

Social network analysis is the study of structure, because the social network approach is grounded in the intuitive notion that the patterning of social ties in which actors are embedded has important consequences for those actors. This is the most important feature of SNA. Early structural intuitions is seen as come from sociologists including Auguste Comte, Ferdinand Tönnies, Emile Durkheim, Sir Herbert Spencer. Freeman argued [8] that these early sociologists all tried to specify the different kinds of social ties that link individuals in different forms of social collectivities. Thus, since they were all concerned with social linkages, they all shared a structural perspective.

If structural intuition is looked as the driving force of social network analysis, we could say that SNA is grounded in systematic empirical data, especially relational, or network data. Relational data is different with the traditional, attribute, or sector data in that it used to describe and explain the relationship between two or more social actors (individuals i and j, for example), while attribute data describe and explain the relationship between two or more attributes of a single social actor (individuals i or j, for example). Tables 1.1 and 1.2 show relational data and attribute data, respectively.

In the context of relational data, as Table 1.1 shows, researcher explores the structure of all the social actors (i.e., i_1, j_1, i_2, j_2 and other nodes), and the mathematical formula could be represented as $S_{ij} = f(ij)$. However, when dealing with attribute data, researcher investigates correlation, causal, mediated, and other relationships

Table 1.1 Relational data

	i_1	j_1	i_2	j_2	...
i_1	–	S_{j1i1}	S_{i2i1}	S_{j2i1}	S_{*i1}
j_1	S_{i1j1}	–	S_{i2j1}	S_{j2j1}	S_{*j1}
i_2	S_{i1i2}	S_{j1i2}	–	S_{j2i2}	S_{*i2}
j_2	S_{i1j2}	S_{j1j2}	S_{i2j2}	–	S_{*j2}
...	S_{i1*}	S_{j1*}	S_{i2*}	S_{j2*}	–

Table 1.2 Attribute data

Y	x_1	x_1	x_3	x_4	...
1	...	...	...	...	...
2	...	...	...	...	...
3	...	...	...	...	...
...	...	...	...	...	...
N	...	...	...	...	...

between different variables (i.e., dependent and independent variables, Y and x_1, x_2, x_3, x_4 and other variables in Table 1.2). Y = f(x) is mathematical formula in this context.

The third prominent feature of SNA is it draws heavily on graphic imagery. In the field of SNA, researchers use points to represent social actors and lines to represent linkages among them. There are two types of graphs: directed and undirected graphs. Directed graph consists of a set of nodes and a set of links (also called arcs or edges). A link e, is an ordered pair (i, j) representing a connection from node i to node j. Node i is called the initial node of link e, $i = \text{init}(e)$, and node j is called the final node of the link: $j = \text{fin}(e)$. If the direction of a link is not important, or equivalently, if existence of a link between nodes i and j necessarily implies the existence of a link from j to i, we say that this network is an undirected graph. A path from node i to node j is a sequence of distinct links (i, u_1), $(u_1, u_2), \ldots$, (u_k, j). The length of this path is the number of links (here $k + 1$). An undirected graph can be represented by a symmetrical matrix $M = (m_{ij})$, where m_{ij} is equal to 1 if there is an edge between nodes i and j, and m_{ij} is 0 if there is no direct link between nodes i and j.

Table 1.3 shows an imaginary relational data matrix. There is a line connecting node A and node B. It is noticeable that this link is an ordered relation. Here A is the initial node of link, while B is the final node. Using directed graph to describe this data matrix, we will get Fig. 1.1.

However, graph showed in Fig. 1.1 is rather primitive, because it only consists of six nodes and seven links. Figure 1.2 describes a more complex graph in a study of China inter-provincial interactions conducted by Jonathan Zhu [6]. In this context, there are 32 nodes (i.e., provinces) and numerous links between any pair of two nodes. There links describe information and population interactions in China mainland. It is clear that Beijing and Shanghai, the biggest cities in China, occupy the centric position in the graph.

Table 1.3 An imaginary relational data matrix

	A	B	C	D	E	F
A	–	1	1	0	0	0
B	0	–	1	1	0	0
C	1	1	–	0	0	0
D	0	0	0	–	1	1
E	0	0	0	0	–	1
F	0	0	0	1	1	–

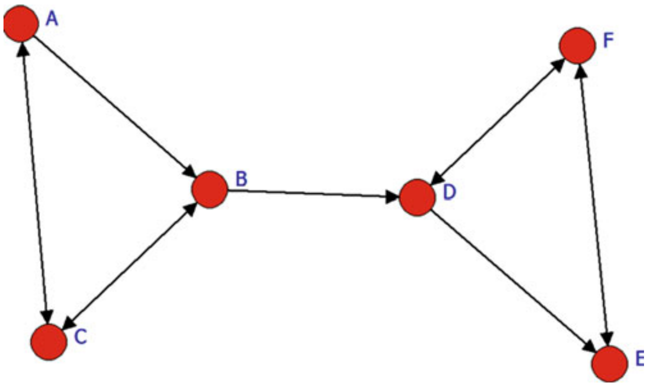

Fig. 1.1 Directed graph of relational data matrix in Table 1.3

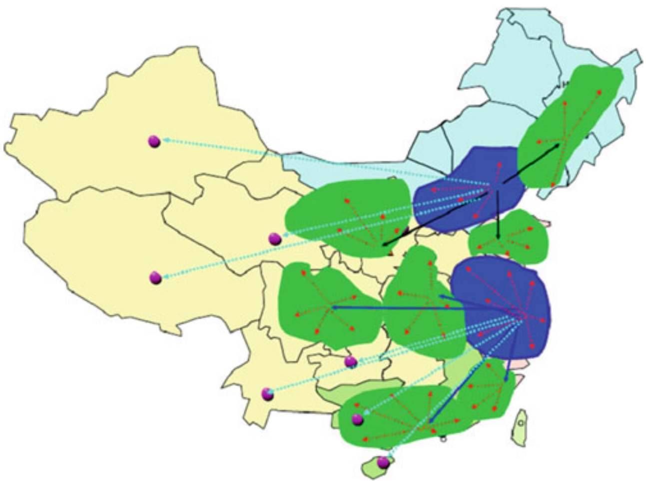

Fig. 1.2 China inter-provincial interactions

Figure 1.3 shows a more complex graph regarding the co-citation network in the research field of media economics [12]. In this graph, it is easy to know that the following documents, such as *Picard_89, Albarran_96, Owen/Wildman_92, Scherer_73/90, Litman_79, Lacy_89,Bagdikian_83/00*, and so on, occupy more

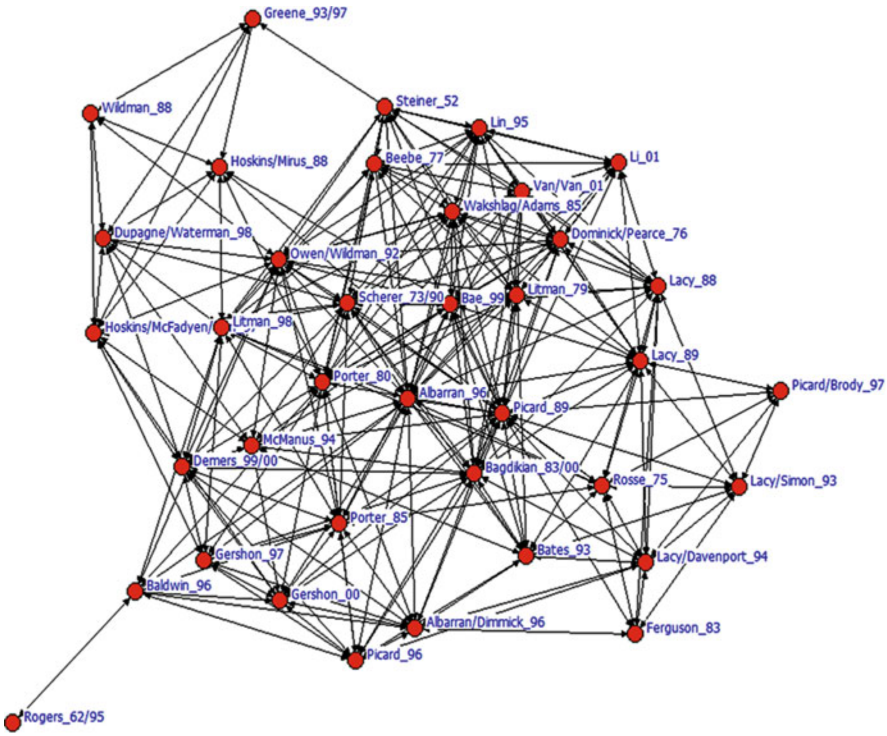

Fig. 1.3 Co-citation network in the research field of media economics

central positions in the network, indicating that they are more important than others. Reflected by the number of ties in the graph, we could see that there are more links point to them.

Freeman argued [8] that unlike many other approaches to social research, network analysis has consistently drawn on various branches of mathematics both to clarify its concepts and to spell out their consequences in precise terms. Thus, the usage of mathematical and/or computational models, is an important feature of SNA. This is the fourth feature of SNA.

1.3 The Development of Social Network Analysis: A Brief History

As stated above, early sociologists in the late 1800s, including Émile Durkheim and Ferdinand Tönnies, are precursors of social network theory. Tönnies argued that social groups can exist as personal and direct social ties that either link individuals who share values and belief or impersonal, formal, and instrumental social links.

Durkheim gave a non-individualistic explanation of social facts arguing that social phenomena arise when interacting individuals constitute a reality that can no longer be accounted for in terms of the properties of individual actors. He distinguished between a traditional society – "mechanical solidarity" – which prevails if individual differences are minimized, and the modern society – "organic solidarity" – that develops out of cooperation between differentiated individuals with independent roles [8]. Georg Simmel, writing at the turn of the twentieth century, was the first scholar to think directly in social network terms. His essays pointed to the nature of network size on interaction and to the likelihood of interaction in ramified, loosely–knit networks rather than groups [13].

Social network as a relatively separate academic concept generated in 1920s–1930s in the research filed of anthropology in Britain. Anthropologist Roger Brown was the first researcher who used the term social network, implying that social structure is similar with a network and that interpersonal communication among individuals resembles relationship between a node and another nesting in the network [14].

One of the line of social network analysis could be traced back to *Sociomery Method* created by social psychologist Jacob Levy Moreno in 1930s, and this method paved the way for quantitative analysis in social network approach. In the 1930s, Moreno pioneered the systematic recording and analysis of social interaction in small groups, especially classrooms and work groups. According to Freeman [8], in Moreno's 1934 book, he used the term "network" in the sense that it is used today. Freeman further pointed out that by 1938, then, the work of Moreno – with the help of Jennings and Lazarsfeld—had displayed all four of the features that define contemporary social network analysis [8].

The second line is a Harvard group led by W. Lloyd Warner and Elton Mayo, which exhibited research effort that focused on the study of social structure in the late 1920s. However, the Harvard effort never "took off" in Freeman's eye, because it never provided a general model for a structural paradigm. As a matter of fact, the efforts at Harvard group are almost never recognized in historical reviews of social network analysis [8].

The 1940s–1960s is called by Freeman as the Dark Ages in the history of the development of SNA. In this period, there was no generally recognized approach to social research that embodied the structural paradigm. Social network analysis was still not identifiable either as a theoretical perspective or as an approach to data collection and analysis [8].

In the 1960s–1970s, a growing number of scholars worked to combine the different tracks and traditions. One large group was centered around Harrison White and his students at Harvard University: Ivan Chase, Bonnie Erickson, Harriet Friedmann, Mark Granovetter, Nancy Howell, Joel Levine, Nicholas Mullins, John Padgett, Michael Schwartz and Barry Wellman. Freeman called it as the Renaissance of SNA at Harvard. The Harvard school published so much important theory and research focused on social networks that social scientists everywhere, regardless of their field, could no longer ignore the idea. By the end of the 1970s, then, social network analysis came to be universally recognized among social scientists.

According to Freeman's statistics, Harrison White group at Harvard University was not the only ones who could lay claim to the social network approach. On the contrary, in the 40-year period from the late 1930s through the late 1970s, there were at least 17 research groups or centers adopted a general social network perspective. Of course, the developments at these groups or centers were not all independent of one another. Those that emerged later undoubtedly drew on the work of at least some of the earlier efforts [8].

Table 1.4 lists some founders and the most prominent researchers of SNA from 1940s through 1970s. However, it can't provide more information regarding the internal structure of these founders and researchers, just like the traditional social research could not discover the interaction patterns of social actors. Figure 1.4 describes the influences of parts of the above founders and researchers. When introducing the basic concepts in the next part of this chapter, we will give more detailed explanations about this figure.

Table 1.4 Noticeable research groups in the development of SNA: 1930s–1970s

Time period	Group leader	Known researchers	University/institution
1930s–1940s	Kurt Lewin, John French, Alex Bavelas	Dorwin Cartwright, Leon Festinger, Duncan Luce	University of Iowa, MIT, University of Michigan
Mid 1940s	Charles P. Loomis	Leo Katz, Charles Proctor, T. N. Bhargava	Michigan State College
Late 1940s	Lévi–Strauss	–	University of Chicago
Early 1950s	Torsten Hägerstrand	–	Lund University
Early 1950s	Nicolas Rashevsky	Walter Pitts, Herbert D. Landahl, Hyman G. Landau, Anatol Rapoport	University of Chicago
Mid 1950s	Paul Lazarsfeld, Robert K. Merton	James S. Coleman, Elihu Katz, Herbert Menzel, Peter Blau, Charles Kadushin	Columbia University
Mid 1950s	Everett M. Rogers	George Barnett, James Danowski, Richard Farace, Peter Monge, Nan Lin, William Richards, Ronald Rice	Iowa State University, Michigan State University
Mid 1950s	Alfred Reginald Radcliffe–Brown, Max Gluckman	John Barnes, John Barnes, J.Clyde Mitchell, Elizabeth Bott, Sigfried Nadel	Manchester University, London School of Economics
Late 1950s	Karl Wolfgang Deutsch, Ithiel de Sola Pool	Fred Kochen	MIT
Late 1950s	Linton C. Freeman, Morris H. Sunshine	Thomas Fararo, Sue Freeman	Syracuse University
Early 1960s	Claude Flament	–	Paris Sorbonne University

(continued)

Table 1.4 (continued)

Time period	Group leader	Known researchers	University/institution
Mid 1960s	Edward O. Laumann	Stephen Berkowitz, Ronald Burt, Joseph Galaskiewicz, Alden Klovdahl, David Knoke, Peter Marsden, Martina Morris, David Prensky, Philip Schumm	University of Michigan
Late 1960s	Peter Blau, James A. Davis	–	University of Chicago
Late 1960s	Robert Mokken	Jac Anthonisse, Frans Stokman	
1960s–1970s	Harrison Colyer White	Peter Bearman, Paul Bernard, Phillip Bonacich, Ronald L. Breiger, Kathleen M. Carley, Ivan Chase, Bonnie Erickson, Claude S. Fischer, Mark Granovetter, Joel Levine, Siegwart M. Lindenberg, Barry Wellman, Christopher Winship	Harvard University

When the concept of social network has been recognized by more and more researchers, more contributions have been made in research methodology: more and more measurement, data collection and analysis technologies were created and developed to understand social construe and relationships better, and these in turn advanced social network research.

In the 1980s, a number of sociologists began to use SNA as analytical technique to examine social and economic phenomena. In mid 1980s, Mark Granovetter, proposed the concept of "embedded-ness", guiding SNA approach into the mainstream social research field again. Granovetter argued that the operation of economics is embedded in social structure, however, the core social structure is individuals' social networks. In early 1980s, Granovetter formulated marketing network theory in his well known article "*Where Do Markets Come From?*" [15].

After 1990s, SNA has been gradually associated with social capital, drawing scholars' attention from the field of sociology, politics, economics, communication science, and other disciplines. Ronald Burt's book *Structural Holes* is representative work of this period. Burt argues that social capital has not relationship with the strength of ties, but with the existence of structural holes. Lin Nan, another well known sociologist who proposed Social-resource theory, studied SNA from the social capital perspective. In the next part of this chapter, when comes to theorization of SNA research design, the author will discuss Burt and Lin more.

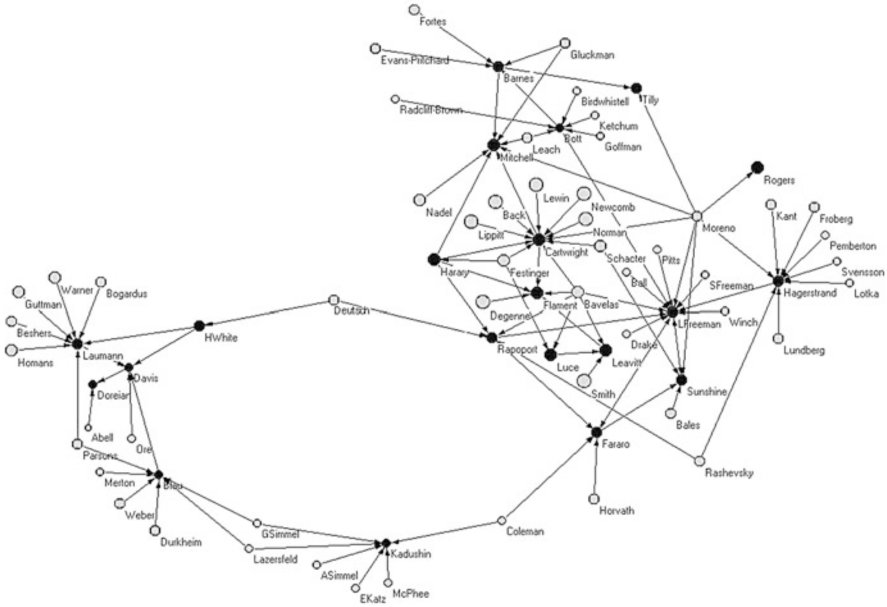

Fig. 1.4 Influences on some founders of SNA [8, p.131]

1.4 Basic Concepts of Social Network Analysis

Social network is formally referred to as a set of social actors, or nodes, members that are connected by one or more types of relations. To understand networks and their participants, researchers should evaluate the location of actors in the network. To measuring location of each node, one should use the concept centrality and other related concepts. Through empirical measurement of a network, one will find various roles and groupings in a network – who are the connectors, mavens, leaders, bridges, isolates? where are the clusters and who is in them? who is in the core of the network? And, who is on the periphery?

In the following of this part, important concepts, such as ties, density, centrality, cliques and other concepts will be explained.

1.4.1 Ties

Ties or links connect two and more nodes in a graph. Many human behaviors, such as advice seeking, information-sharing, and lending money to somebody are directed ties while co-memberships are examples of undirected ties. Directed ties may be reciprocated, as would be the case for two people who visit one another, or they may exist in only one direction as when only one gives emotional support to